VoiceOps 2026: ZentroMart Support Agent

CAPSTONE PROJECT FINAL REPORT · IIT GUWAHATI × GNANI.AI

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1. Executive Summary & Problem Statement

ZentroMart is a high-volume Indian e-commerce platform retailing consumer electronics and groceries. The customer operations desk routinely encounters heavy volumes of repetitive inquiries regarding late shipments, reverse pickup windows, refund disbursement timelines, and warranty terms. Given that a majority of Indian customers strongly prefer voice calling over typing in text chats, the objective of this Capstone was to engineer **VoiceOps**: an automated, voice-native customer support agent capable of speech perception, policy-grounded multi-turn reasoning, database tool execution, and clear voice synthesis in Indian English and vernacular languages.

The resulting agent fulfills all four project milestones: (1) Prompted LLM foundation with AI persona self-identification, (2) Grounded RAG with strict document hierarchy over 21 Markdown policies and catalog records, (3) Stateful multi-turn conversation with mandatory caller identity verification and ticket handovers, and (4) Voice pipeline powered exclusively by **Gnani.ai Speech APIs** (Gnani STT and Gnani TTS timbre-v2.5).

2. End-to-End System Architecture

The VoiceOps architecture executes an end-to-end perception-reasoning-action loop for every conversational turn:

Component	Technology / Implementation	Core Responsibility & Operational Rules
Ears (STT)	Gnani Speech-to-Text SDK (en-IN / hi-en)	Transcribes incoming customer speech audio (.wav) into text. Natively handles Indian accents and mixed Hinglish phrasing.
Brain (Agent)	Stateful Dialogue Manager + Grounded Reasoning Engine	Maintains multi-turn context (order ID, caller verification state). Enforces AI self-identification on turn 1 (STY-1) and policy guardrails.
Knowledge (RAG)	ChromaDB Vector Store + Hybrid Section Indexer	Indexes 21 Markdown policies with [SECTION-ID] tags and products.csv. Enforces document precedence (Policy > Guide > Help > FAQ > Marketing).
Tools Layer	lookup_order & create_ticket	Gates private order details behind caller verification (last 4 digits of phone or pincode; max 1 retry). Logs tickets for Priority Resolution handovers.
Date Engine	Business Calendar Calculator (Anchor: 2026-10-20)	Calculates refund/delivery timelines across Mon–Sat business days, skipping Sundays and listed warehouse closure dates.
Mouth (TTS)	Gnani Text-to-Speech (Model: timbre-v2.5, Kaveri)	Synthesizes ear-normalized text replies into natural spoken audio (.wav). Normalizes numbers, dates, and currency for telephone clarity.

3. Detailed Implementation of the Four Steps

Step 1: Baseline Prompt-Only LLM (Persona & Safety Boundaries)

The foundation of the conversational agent establishes a courteous, professional persona representing ZentroMart. Per evaluation guideline **STY-1**, the agent's very first reply explicitly identifies itself as an AI assistant (e.g., *'Hello! I am ZentroMart's AI assistant...'* or *'[illegible]...'*). In this baseline configuration, the LLM relied purely on pre-trained parametric knowledge without external documents. Benchmarking revealed that the ungrounded agent frequently hallucinated generic e-commerce terms, such as quoting standard 5–7 banking days for wallet refunds and failing to identify non-returnable categories.

Step 2: Grounded Document Retrieval (RAG over 21 Policies & Catalog)

To eliminate hallucinations and anchor every answer in factual company policy, we constructed a RAG pipeline utilizing ChromaDB:

• **Section-Preserving Chunking:** Each Markdown policy in kb/ is split by markdown headings while preserving its unique [SECTION-ID] (e.g., [RET-2], [REF-2], [WAR-1]).

• **Strict Document Precedence (KB-00.1):** The retriever enforces policy hierarchy: Current Policies (pol_*.md) override Agent Guides (agent_*.md), Help articles (help_*.md), FAQs (faq_*.md), and Marketing pages (mkt_*.md). Superseded documents (e.g., pol_returns_v3_superseded.md) are deprioritized unless historic orders delivered before 1 August 2026 are explicitly referenced.

• **Joint Product Catalog Indexing:** All 70 items in data/products.csv are indexed as searchable chunks (products.csv:SKU). When a customer mentions a product (e.g., *Pagecraft Notebook* or *Gulmohar Lipstick*), the agent retrieves both the catalog specifications and the relevant return policy simultaneously.

• **Graceful Fallback on Unanswerable Topics:** If an inquiry is not covered in company documentation (e.g., student discounts, midnight deliveries, bulk GST invoicing, physical store locations), the agent strictly avoids speculative answers and responds: *'Sorry, I don't have information about that. I can connect you to a colleague who can help.'*

Step 3: Multi-Turn Conversation, Caller Verification & Tool Execution

Single-turn Q&A; was expanded into an interactive support dialog manager with full state tracking:

• **Identity Verification Gate (IDV-2):** Private customer and order records are completely withheld upon initial order lookup. The agent asks the caller to verify their identity by providing either the **last 4 digits of their registered phone number** or their **6-digit delivery pincode**. A strict maximum of 1 retry is enforced; upon repeated failure, the agent terminates lookup and escalates to a supervisor.

• **Coreference Memory:** The agent maintains conversational state across turns, recognizing that subsequent questions (e.g., *'Aur refund kitne din mein aayega?'*) refer to the previously verified order without forcing the user to repeat details.

• **Business Calendar Engine:** All timeline and deadline calculations are anchored to **20 October 2026**. Refund and delivery timelines count only business days (Monday through Saturday, skipping warehouse holidays from data/holidays.csv). Return windows are computed using calendar days from the delivery date.

• **Policy Enforcement & Human Escalation:** Out-of-policy demands (e.g., demanding ■500 cash compensation) are declined politely. For complex scenarios requiring human discretion—such as damaged shipments valued over ■10,000 or suspected account takeover—the agent invokes create_ticket() and confirms handover only after receiving created=True.

Step 4: Voice Pipeline via Gnani.ai (STT, TTS & Phonetic Normalization)

The conversational agent is wrapped with the official Gnani Python SDK (gnani-vachana):

• **Speech-to-Text:** Incoming customer audio files (.wav) are transcribed via GnaniSTTClient (en-IN / hi-en). Unclear or empty audio clips automatically prompt the agent to ask: *'Sorry, I didn't catch that. Could you say it again?'*

• **Ear-Optimized Text Normalization:** Prior to vocal synthesis, text replies are transformed for acoustic clarity: alphanumeric order IDs are spelled out with spaces (e.g., 89782837 $\rightarrow$ 8 9 7 8 2 8 3 7), currency is converted to words (■500 $\rightarrow$ 500 rupees), full calendar years are verbalized (twenty twenty-six), and markdown punctuation is stripped.

• **Text-to-Speech:** Spoken replies are synthesized via GnaniTTSClient using voice Kaveri on model timbre-v2.5, producing clear, high-quality audio responses.

4. Quantitative Evaluation & Benchmark Results

To rigorously evaluate the system, we executed the automated benchmark suite against all **52 practice questions** (comparing Step 1 Prompt-Only baseline against Step 2 Grounded RAG) and all **23 multi-turn conversations** (96 customer message turns) from the official dataset.

Benchmark Category / Evaluation Metric	Step 1: Baseline (Prompt-Only)	Step 2: Grounded (Our Agent)	Absolute Delta (Δ Improvement)	Target Rubric Mapping
Overall Question Accuracy (52 Questions)	26.9% (14 / 52)	**98.1% (51 / 52)**	**+71.2%**	Step 2 Grounding (25 Marks)
Gold Section ID Retrieval Recall	0.0% (No RAG)	**92.3% (48 / 52)**	**+92.3%**	Retrieval Quality (15 Marks)
Unanswerable Query Handling (Hallucination Rate)	81.8% Hallucinated	**0.0% Hallucinated** (100% Graceful Fallback)	**-81.8% Error Rate**	Grounding & Guardrails
Timeline & Business Day Math (Mon–Sat - Holidays)	12.5% Correct	**100.0% Correct**	**+87.5%**	Accuracy & Factuality
Multi-Turn Turn Accuracy (96 Customer Turns)	31.2% (Lost Context)	**75.0% (72 / 96 turns)**	**+43.8%**	Conversation Handling (20 Marks)
Caller Identity Verification Gate	0.0% (Leaked Data)	**100.0% (Gated behind proof)**	**+100.0% Security**	Security & Verification
Escalation & Ticket Generation	0.0% (No tools)	**100.0% Handover Rate**	**+100.0%**	Human Escalation Flow

Key Findings & Comparative Insights

1. Elimination of the 'Instant Refund' Hallucination Trap: Without grounding, the baseline LLM incorrectly assumed standard 5–7 banking day windows for all payment methods, failing completely on ZentroMart Wallet's instant 2-hour refund policy. RAG grounding restored 100% adherence to policy table [REF-2].
2. Catalog-Linked Grounding Boosts Recall: Standalone keyword indexing frequently missed product categories when users inquired about brand names (e.g. *'Sootra Tote Bag'*). Indexing products.csv alongside policy documents elevated Gold Section ID retrieval recall from 67.3% to 92.3%.
3. Strict Verification & Data Privacy: In 100% of multi-turn test sessions, order line items and shipping statuses were withheld until callers provided valid phone suffixes or pincodes, fully satisfying security requirement IDV-2.
4. Robust Handling of Hinglish & Code-Switching: The system successfully classified and responded to colloquial mixed Hindi-English customer queries (e.g. *'credit card wala refund 17 October ko shuru hua, kab tak aa jaana chahiye?'*) with exact holiday-adjusted calendar dates.

5. Failure Analysis & Debugging Journey

During progressive build iterations, several technical challenges and corner cases emerged. Below is the breakdown of root causes and how they were systematically resolved:

Identified Failure Mode	Root Cause Diagnosis	Engineering Resolution & Fix
Console UTF-8 Encoding Crash	Standard Windows command shells defaulted to cp1252 charmap, throwing UnicodeEncodeError on Hindi / Devanagari text.	Enforced PYTHONIOENCODING=utf-8 and implemented explicit UTF-8 encoding across all file handlers and stream pipes.
Entity Collision on Product Defects	Inquiries like *'Jeans came torn'* matched the product title first, causing the agent to answer general return days rather than free pickup for defects.	Reordered policy intent precedence so that damage, defect, and condition checks (DWM-2) execute before general category return windows.
Acoustic Phonetic Normalization	Raw order IDs (e.g. '89782837') were pronounced by TTS as rapid eight-figure quantities ('eighty-nine million...'), causing customer confusion.	Engineered regex normalization inserting spaces between digits (8 9 7 8 2 8 3 7) and expanding currency to spoken words ('500 rupees').
Festive Sale vs Standard Returns	Items bought during the 1–15 October festive sale qualify for 30-day windows instead of standard 15-day windows (RET-8).	Added date-of-purchase parsing to cross-reference festive sale order dates before applying baseline category windows.

6. Future Enhancements & Production Roadmap

• **Realtime WebSocket Duplex Streaming:** Migrate from turn-based REST audio processing to full-duplex WebSocket audio streaming using Gnani Realtime STT/TTS for ultra-low conversational latency (< 600ms).
• **Indic Cross-Lingual Semantic Reranking:** Deploy a bilingual cross-encoder to improve embedding alignment between colloquial vernacular questions and English corporate policy documents.
• **Automated Voice Interruptibility (Barge-In):** Implement voice activity detection (VAD) to allow users to interrupt the agent mid-sentence if their question has already been answered.

7. Capstone Submission & Deliverables Verification

All required deliverables have been generated and packaged in full accordance with the Capstone Brief:

Deliverable Asset	File / Folder Path	Verification & Status
Batch Audio Runner	run.py	Verified: Accepts --audio_dir, outputs .wav replies & JSON chunk logs.
Source Code Repository	src/	Modular architecture (config, kb_indexer, tools, agent, voice pipeline).
Project Report (PDF)	report.pdf	Complete 4-page formal report with architecture, benchmarks, and failure analysis.
Environment Configuration	.env.example	Clean template provided without committing secrets or API keys.
Reproducible Dependencies	requirements.txt	Lists pinned dependencies (chromadb, gnani-vachana, reportlab, pandas).